

Connected Architectures and Fashion MNIST

The Uncs

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Abstract—This project involved a connected neural networks for classifying Fashion-MNIST images into ten clothing categories. Two network architectures were studied with two learning rates, both with and without dropout regularization being used. This resulted in eight experimental configurations. Each network used ReLU activation functions for the hidden layers, a softmax output layer, categorical cross-entropy loss, and Adam optimizer. The Fashion-MNIST training data were separated into 90% training data and 10% validation data, and the model selection was based on the validation accuracy. The selected network contained two hidden layers made of 400 neurons, a learning rate value of 10^{-3} , and a dropout rate of 0.2. This achieved 90.18% validation accuracy and 88.28% test accuracy. The estimated generalization error was 11.72% with a 95% confidence interval of [11.09%, 12.35%]. The results show that a connected neural networks can classify Fashion-MNIST, however visually similar clothing classes were still difficult to distinguish.

I. INTRODUCTION

Fashion-MNIST [1] contains grayscale images from ten clothing categories and gives a classification problem in which several classes, such as shirts, T-shirts, pullovers, and coats, have similar visual structures.

This project evaluated how network architecture, learning rate, and dropout regularization can affect connected neural network performance on Fashion-MNIST. Two architectures, two learning rates, and two dropout settings produced eight configurations. All of the models used ReLU hidden-layer activations, a softmax output layer, categorical cross-entropy loss, and Adam optimization. The 400–400 model with a learning rate of 10^{-3} and dropout of 0.2 achieved the highest validation accuracy of 90.18% and was selected for final analysis.

II. PROBLEM AND METHODS

The input to the classifier was a 28×28 grayscale Fashion-MNIST image with pixel intensity values are integers ranging from 0 to 255. Each image was assigned to one of ten output classes, which were T-shirt/top, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, or ankle boot. Pixel normalization was performed inside the network by scaling the input values to the interval $[0, 1]$. Class labels were converted into ten-dimensional one-hot vectors, and categorical cross-entropy was used as the loss function.

Two connected architectures were evaluated. The first architecture contained two hidden layers each with 400 neurons, while the second architecture contained only one hidden layer that had 128 neurons. All hidden layers used ReLU activation. Each network used a ten-neuron softmax output layer. The larger network was used to study if an increased capacity

TABLE I
VALIDATION ACCURACY FOR EXPERIMENTAL CONFIGURATIONS

Architecture	LR	Dropout	Val. Acc.
400–400	10^{-3}	0.2	90.18%
400–400	10^{-4}	0.2	90.12%
128	10^{-4}	0.2	89.77%
400–400	10^{-3}	0.0	89.73%
400–400	10^{-4}	0.0	89.65%
128	10^{-3}	0.0	89.37%
128	10^{-3}	0.2	89.28%
128	10^{-4}	0.0	89.22%

improved the classification, while the smaller architecture gave a lower-complexity comparison.

10^{-3} and 10^{-4} learning rates were tested with dropout rates of 0 and 0.2. These produced eight total configurations. Adam was the optimizer. The original training data were separated into 90% training data and 10% validation data. Validation accuracy was used for model selection. Early stopping monitored the validation loss with a patience of four epochs and restored the weights from the best epoch.

III. EXPERIMENTAL RESULTS

The validation performance for all eight of the different configurations studied for this project can be seen on Table I. The 400–400 architecture with a learning rate of 10^{-3} and dropout of 0.2 had the highest validation accuracy of the different configurations, with a value of 90.18%. Figure 1 shows the training and validation loss and accuracy. This shows both convergence and overfitting behaviors.

The selected model had a test accuracy of 88.28% on the 10,000 test examples. The estimated generalization error was 11.72%, the 95% confidence interval was [11.09%, 12.35%], and the equivalent accuracy interval was [87.65%, 88.91%].

Figure 2 shows the normalized confusion matrix for the selected model. Bag, trouser, ankle boot, and sneaker were the most accurate classes, with accuracies of 97.5%, 97.1%, 96.4%, and 95.9% respectively. Shirt was the least accurate class, with an accuracy of only 64.2%, and was often confused with the T-shirt/top class. Pullover and coat also had lower accuracy ratings of 78.7% and 79.9% respectively.

For the 400–400 model at 10^{-3} , dropout reduced the training-validation gap from 92.60%/89.73% to 91.00%/90.18%, suggesting less overfitting. However, the 95% confidence intervals for the test-error differences all contained zero.

Figure 3 shows prediction examples and gradient-times-input saliency maps. Brighter regions show pixels with greater

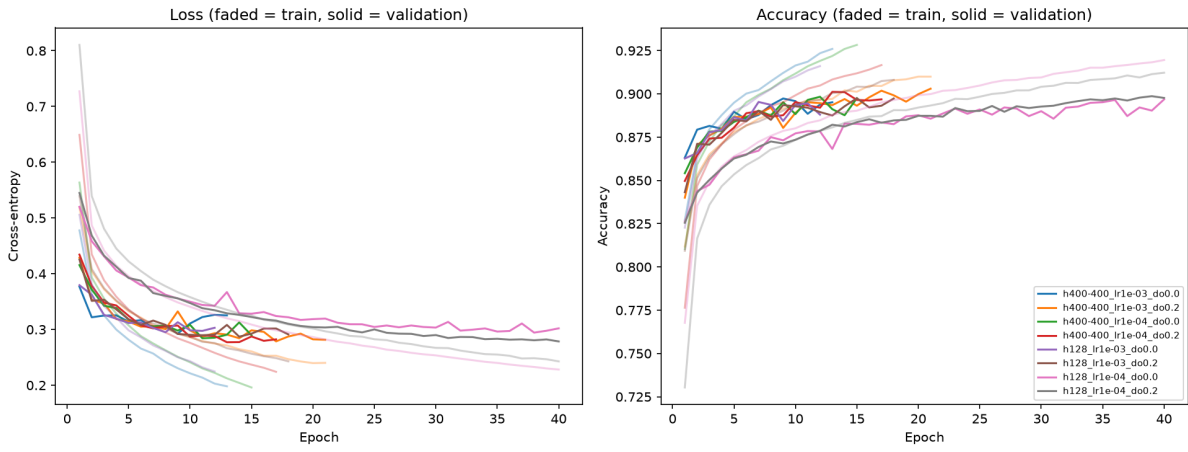


Fig. 1. Training and validation cross-entropy loss and accuracy for all eight experimental configurations. Faded lines represent training results and solid lines represent validation results.

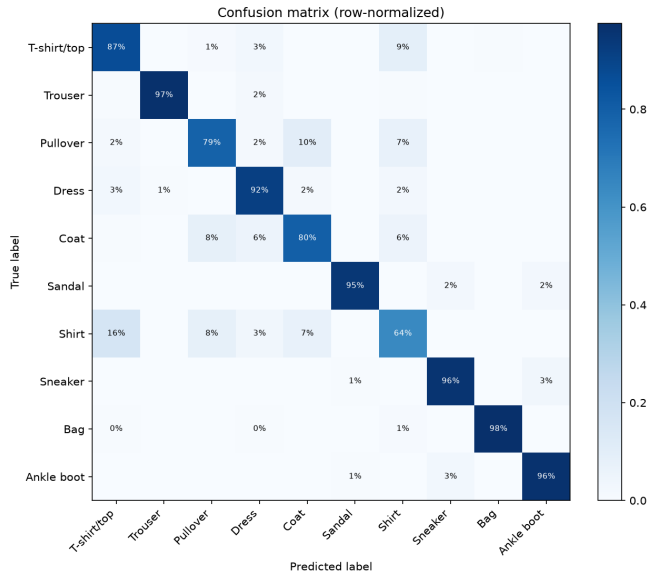


Fig. 2. Row-normalized confusion matrix for the selected model.

impact on the predicted class. Important regions were usually located on the clothing item, especially along boundaries and distinctive features. Misclassifications were mostly found on visually similar classes such as shirts, T-shirts, pullovers, and coats.

IV. DISCUSSION

The larger 400–400 architecture had a higher validation accuracy than the single 128-neuron architecture, which indicates that more model capacity was useful for this specific task. Dropout improved the validation accuracy in several configurations and reduced the difference between training and validation performance for the selected model. The learning rate also affected the performance, however there was no

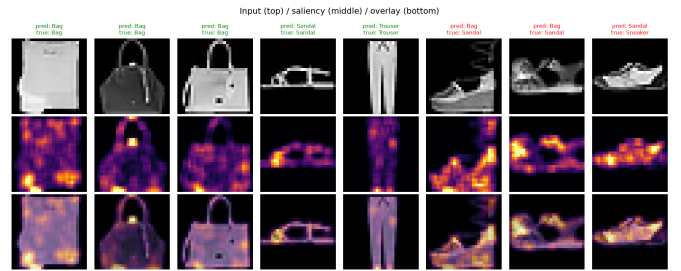


Fig. 3. Example predictions with gradient-times-input saliency maps. Brighter regions indicate pixels with greater influence on the output.

learning rate that consistently was the best for all of the configurations.

One of the main advantages of the connected approach that it is simple. The networks are considerably straightforward for training and evaluation. Also, dropout provides a simple form of regularization. Saliency maps also provide a way to visualize where pixels most impact the individual predictions. One of the disadvantages is that flattening each image removes its unique two-dimensional spatial structure. This likely impacts the accuracy for choosing between similar classes with differences that depend on certain similar shapes.

V. CONCLUSION

Two connected neural network architectures were evaluated on Fashion-MNIST using different learning rates and dropout settings. The 400–400 architecture with a learning rate of 10^{-3} and dropout of 0.2 had the highest validation accuracy of 90.18% and was selected as the best and final model. This model had a test accuracy of 88.28% with an estimated generalization error of 11.72% and a 95% confidence interval of [11.09%, 12.35%]. The results show that architecture size, learning rate, and regularization influence generalization performance. The remaining classification errors are concentrated mostly on visually similar clothing categories.

APPENDIX A OTHER RESULTS

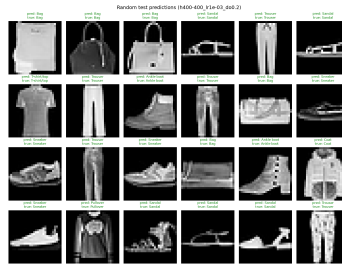


Fig. 4. Random Fashion-MNIST test examples with the predicted and true classes for the selected model.

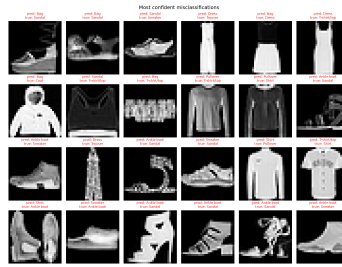


Fig. 5. Test examples relating to the selected model's largest misclassifications.

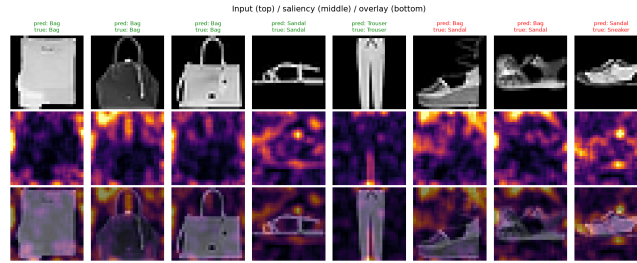


Fig. 6. Example predictions with gradient-based saliency maps. Brighter regions relate to pixels having greater impact on the predicted output.

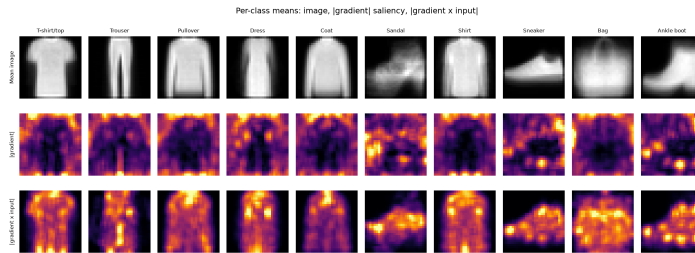


Fig. 7. Mean image and saliency results for each Fashion-MNIST class, including the gradient and gradient-times-input saliency.

REFERENCES

- [1] H. Xiao, K. Rasul, and R. Vollgraf, “Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms,” *arXiv preprint arXiv:1708.07747*, 2017.